

Card Matching Memory Game

Design Document

1. Game Overview

This is a memory puzzle game where players flip cards to find matching pairs. The goal is to clear the board before the time runs out or the player uses too many moves. The game gets harder as you progress, unlocking new levels with bigger grids.

2. Design

I designed the game to be efficient and modular. Instead of creating many different scenes for every level, I used a smart "One-Scene" system.

Menu Scene: The player chooses a level (Easy, Medium, or Hard).

Game Scene: The game reads the choice and automatically builds the correct grid size (e.g., 2x3, 3x4, 5x6 or 6x6) using code.

3. Game Manager

Grid Generation: The game calculates the screen size and spawns cards to fit perfectly, whether it's a small 3x4 grid or a large 5x6 grid.

Game Loop: It handles the Timer, Move Counter, and checks if the player has won or lost.

4. Difficulty & Progression

I created a flexible "Level Data" system. Each difficulty setting has its own rules:

Easy: Small grid (6 cards).

Medium: Medium grid (16 cards).

Hard: Large grid (30 cards).

Difficult: Much Larger grid (36 cards), strict move limit.

When a player wins, the game saves their progress (unlocking the next level) and gives them a Star Rating (1-3 stars) based on how fast they finished.

5. Save System

The game uses Unity's PlayerPrefs to save data. This ensures that even if the player closes the game, their Unlocked Levels and High Scores (Stars) are remembered.

6. Audio & Visuals

Feedback: Cards have a smooth flipping animation.

Audio: There are specific sounds for flipping, matching, mistakes, and winning to make the game feel responsive.